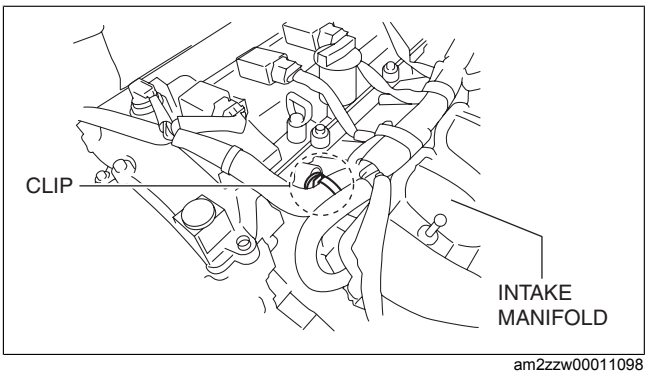
ENGINE DISASSEMBLY/ASSEMBLY [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0110q3800500

Caution
- **When the transaxle is installed, do not suspend the engine. Otherwise, the SST installation area of the cylinder head may be damaged due to excess weight. Only suspend the engine after separating the engine and transaxle.**
- **Applying excessive force (force of 100 N {10.2 kgf, 22.5 lbf} or more) to the electric variable valve timing motor/driver may cause a malfunction. When servicing, be careful not to apply excessive force to the electric variable valve timing motor/driver using other parts or tools.**

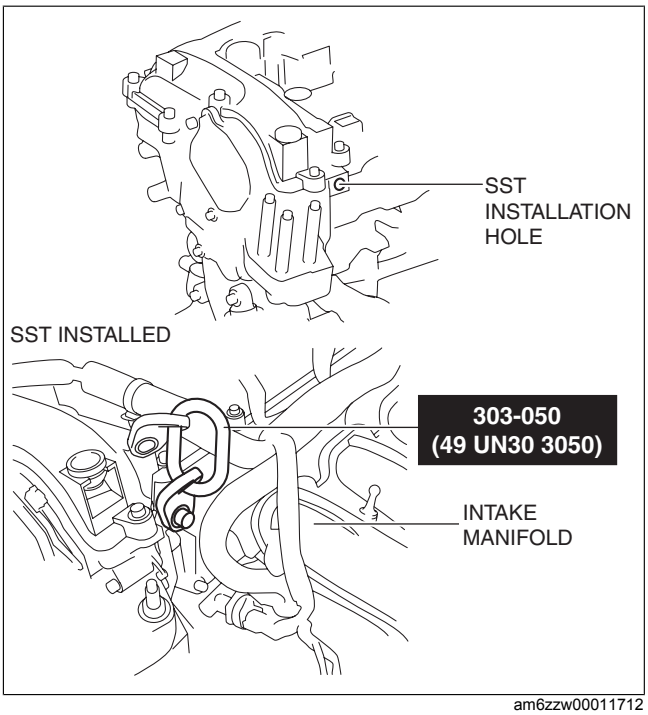
1. To enable to install the **SST**, disconnect the clip shown in the figure and set the wiring harness aside.



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2. Install the **SST** and the following bolt to the position shown in the figure.

Engine front side

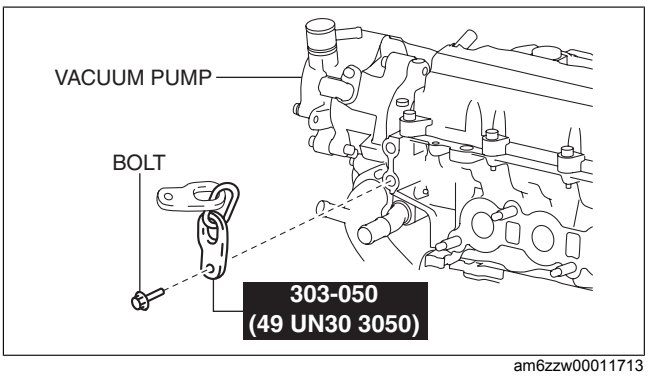


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Engine rear side

Bolt:
part number 99794 1025 or an M10 × 1.25, length 25 mm {0.98 in}

Tightening torque
38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}

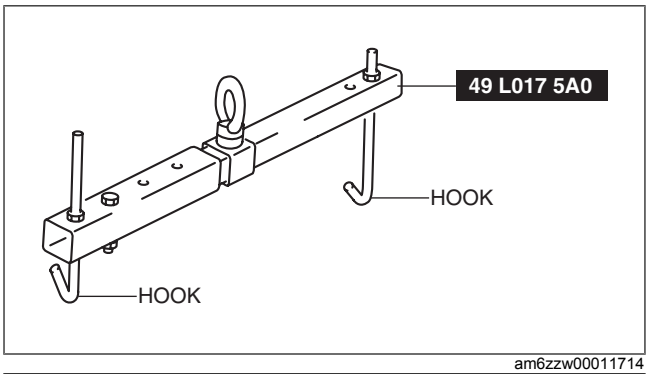


am6zzw00011713

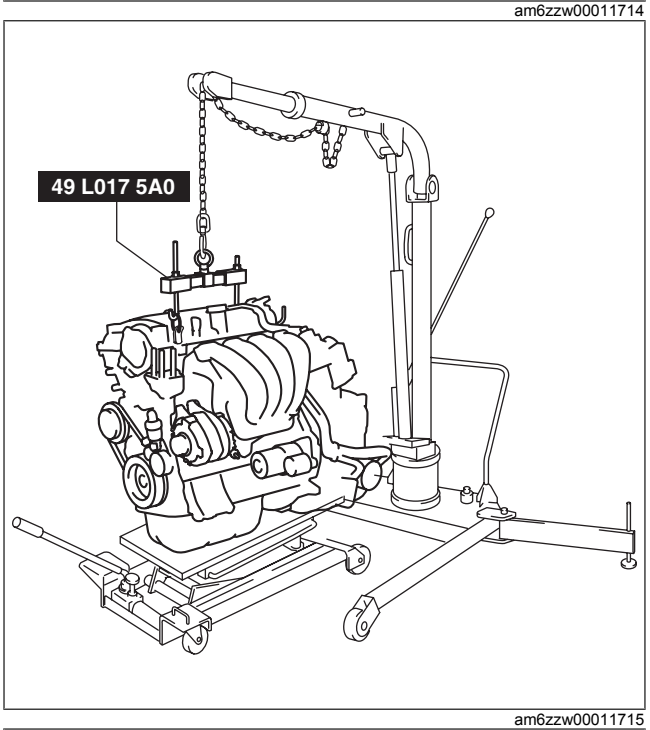
3. Engage the hooks of the **SST (49 L017 5A0)** to the **SST (49 UN30 3050)**.

   Caution
   - **Do not lift up the engine.**

4. To ensure the safety of the work (control engine and transaxle sway), set a hoist as shown in the figure.

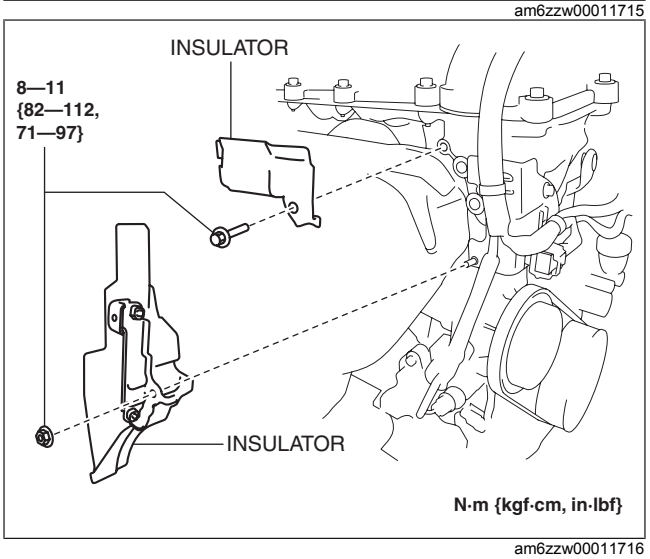


am6zzw00011714



am6zzw00011715

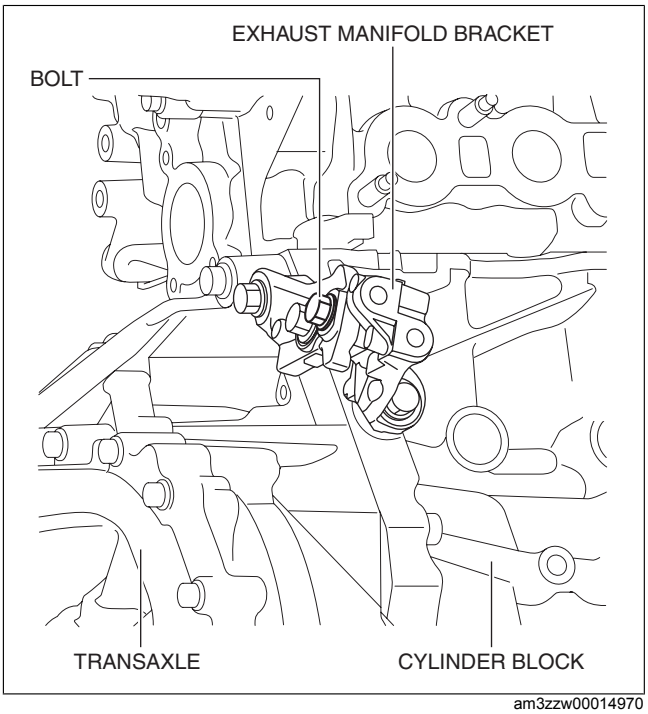
5. Remove the insulator shown in the figure. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
6. Remove the exhaust system. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)



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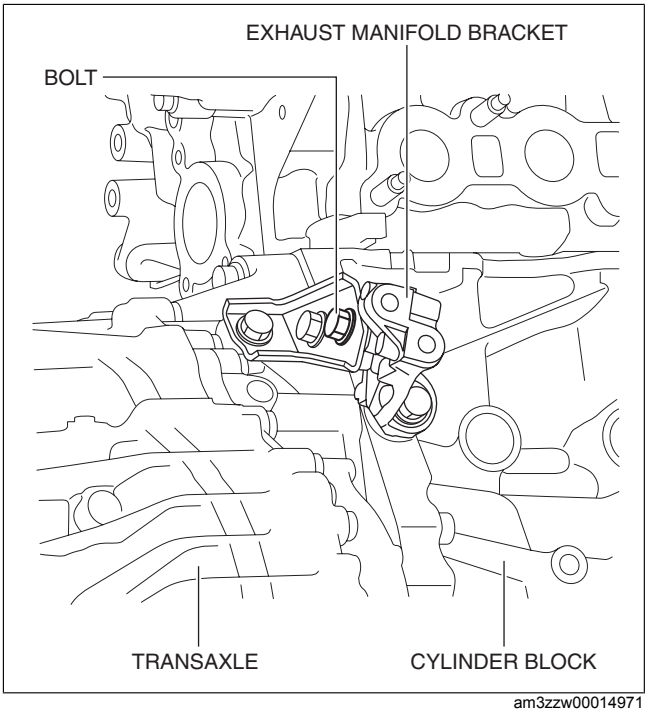
7. Remove the exhaust manifold bracket using the following procedure: (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
   (1) Loosen the bolt shown in the figure.

MTX



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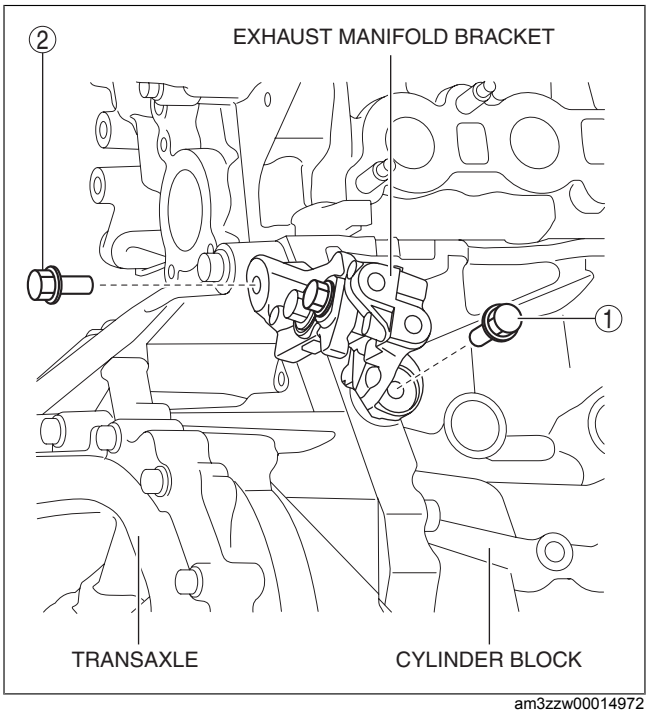
ATX



am3zzw00014971

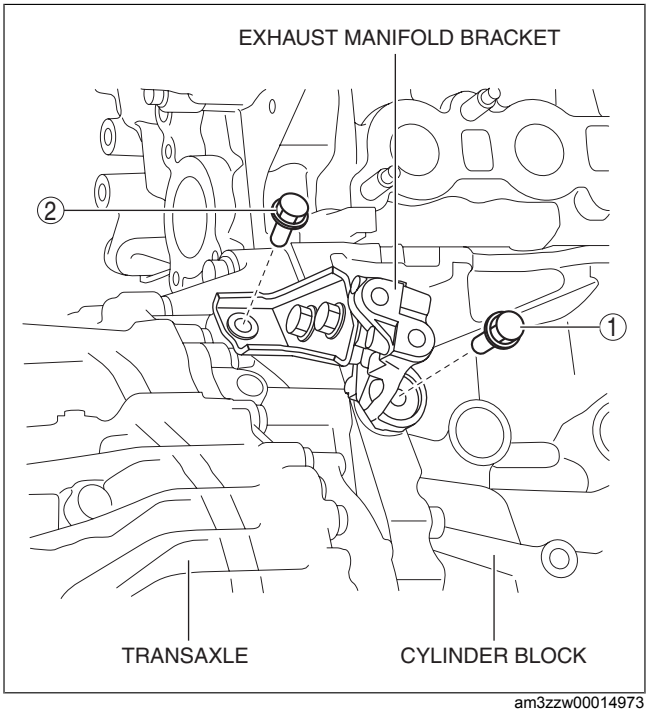
(2) Remove the bolts in the order shown in the figure, then remove the exhaust manifold bracket.

MTX



am3zzw00014972

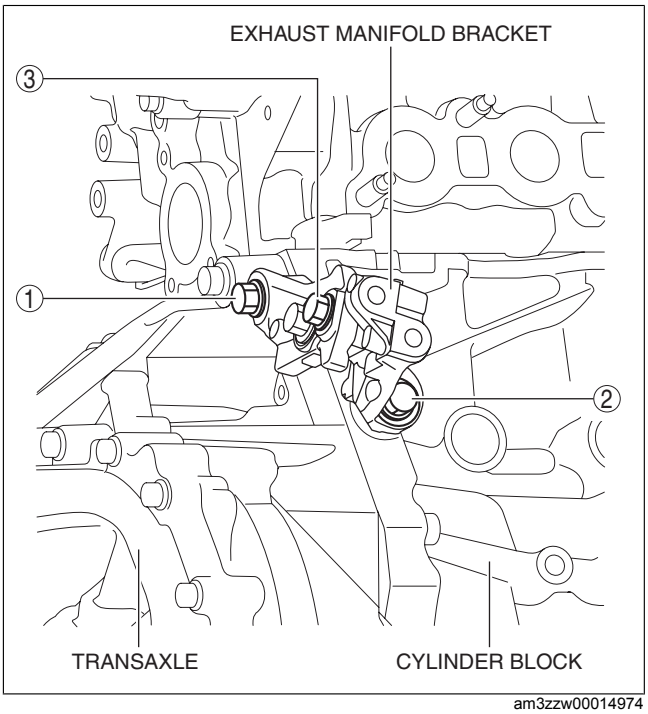
ATX



am3zzw00014973

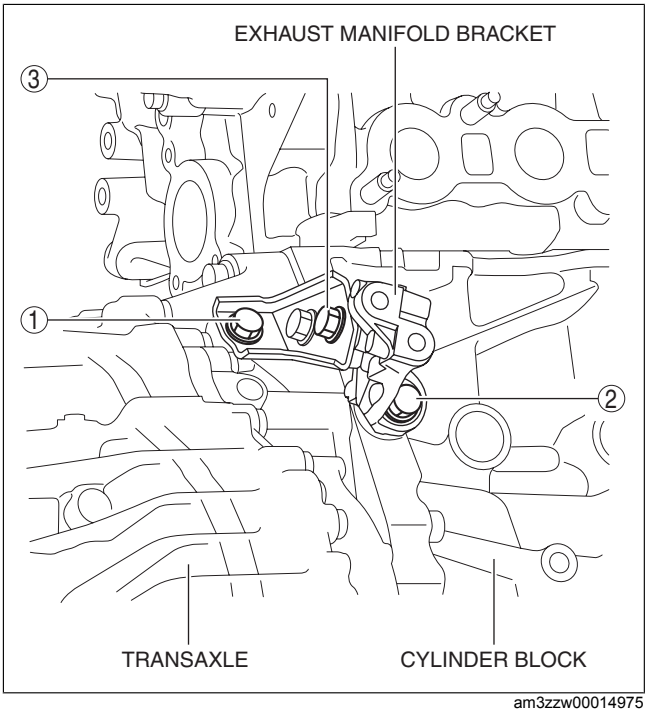
- When installing, tighten the bolts in the order shown in the figure.

MTX



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ATX



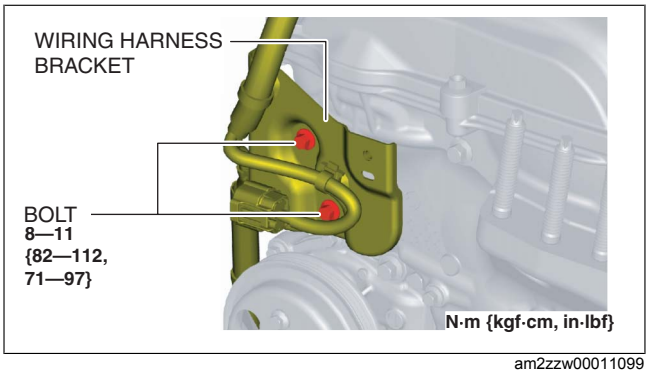
am3zzw00014975

Tightening torque
38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}

8. Remove the bracket No.1 (drive shaft bracket). (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
9. Remove the starter. (See STARTER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
10. Fix the drive plate using the crankshaft pulley lock bolt. (ATX)
11. Remove the torque converter installation nut from the starter installation hole. (See AUTOMATIC TRANSAXLE REMOVAL/INSTALLATION [CW6A-EL].)
12. Disconnect the engine and transaxle, and lower only the engine from the engine lifter. (See MANUAL TRANSAXLE REMOVAL/INSTALLATION [F65M-R](MTX).) (See MANUAL TRANSAXLE REMOVAL/INSTALLATION [F66M-R (SKYACTIV-G 1.5)](MTX).) (See AUTOMATIC TRANSAXLE REMOVAL/INSTALLATION [CW6A-EL] (ATX).)
13. Remove the generator. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)
14. Remove the intake-air system. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
15. Remove the fuel injectors. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
16. Remove the camshaft position (CMP) sensor. (See CAMSHAFT POSITION (CMP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
17. Remove the vacuum pump. (See VACUUM PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
18. Remove the high pressure fuel pump and rear housing. (See HIGH PRESSURE FUEL PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
19. Remove the oil filter. (See OIL FILTER REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

20. Remove the oil pressure switch. (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)) (See OIL PRESSURE INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
21. Remove the engine oil solenoid valve. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system)) (See ENGINE OIL SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
22. Remove the crankshaft position (CKP) sensor. (See CRANKSHAFT POSITION (CKP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
23. Remove the ignition coils. (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)) (See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
24. Remove the ignition coil/ion sensors. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system)) (See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
25. Remove the wiring harness bracket shown in the figure.

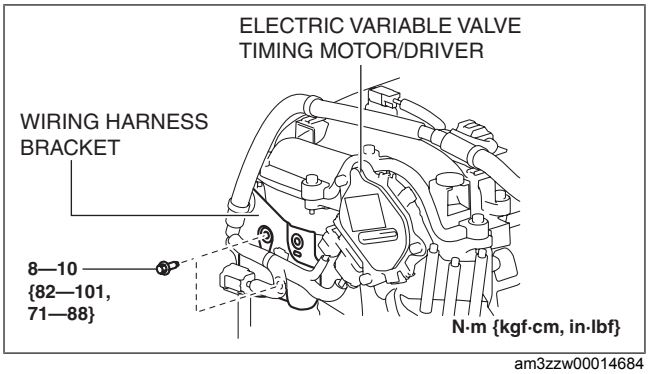
SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)



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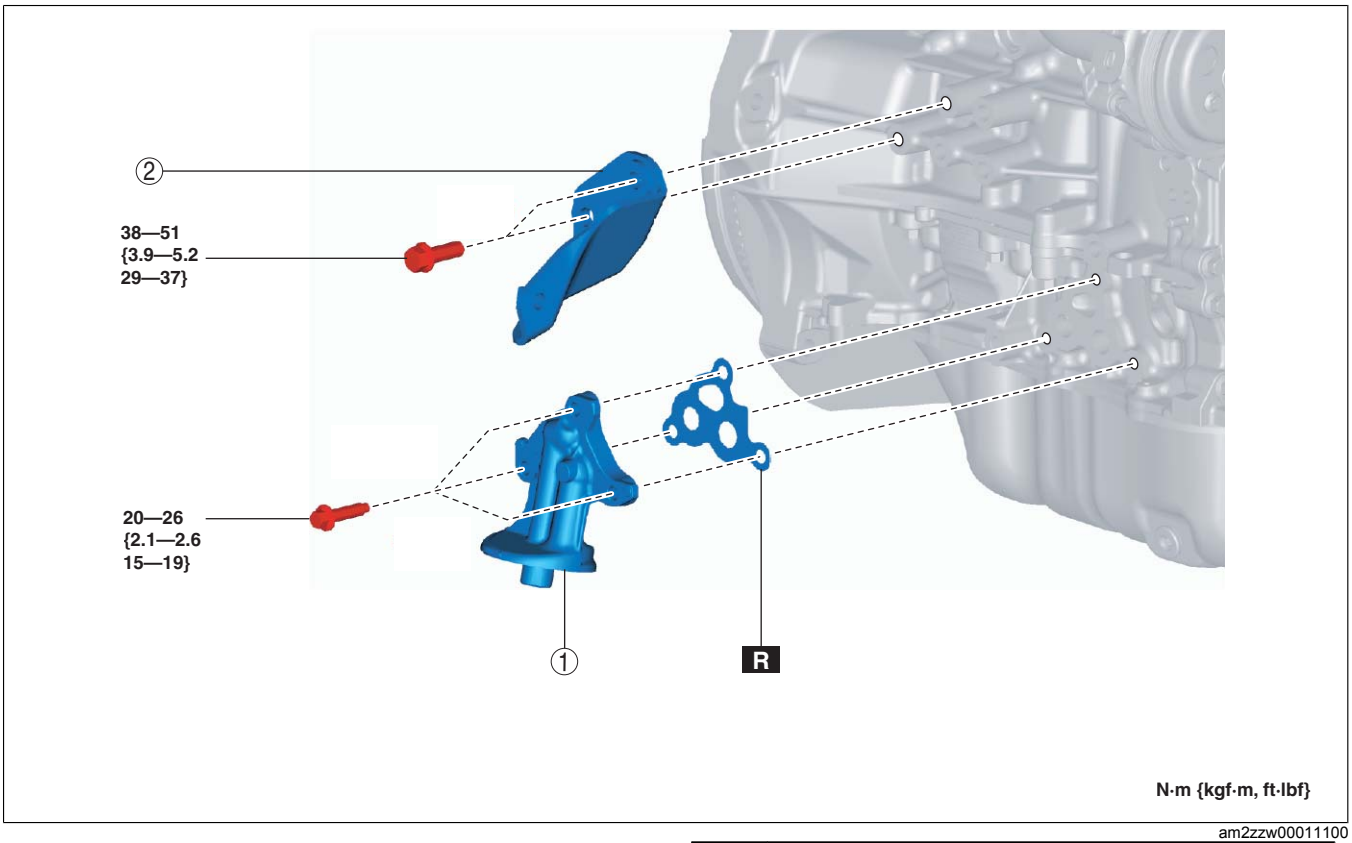
SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

26. Remove the emission harness.
27. Remove the water pump drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
28. Remove the clutch cover and clutch disc. (MTX) (See CLUTCH UNIT REMOVAL/INSTALLATION [F65M-R].) (SeeCLUTCH UNIT REMOVAL/INSTALLATION [F66M-R].)
29. Remove in the order indicated in the table
30. Assemble in the reverse order of disassembly.



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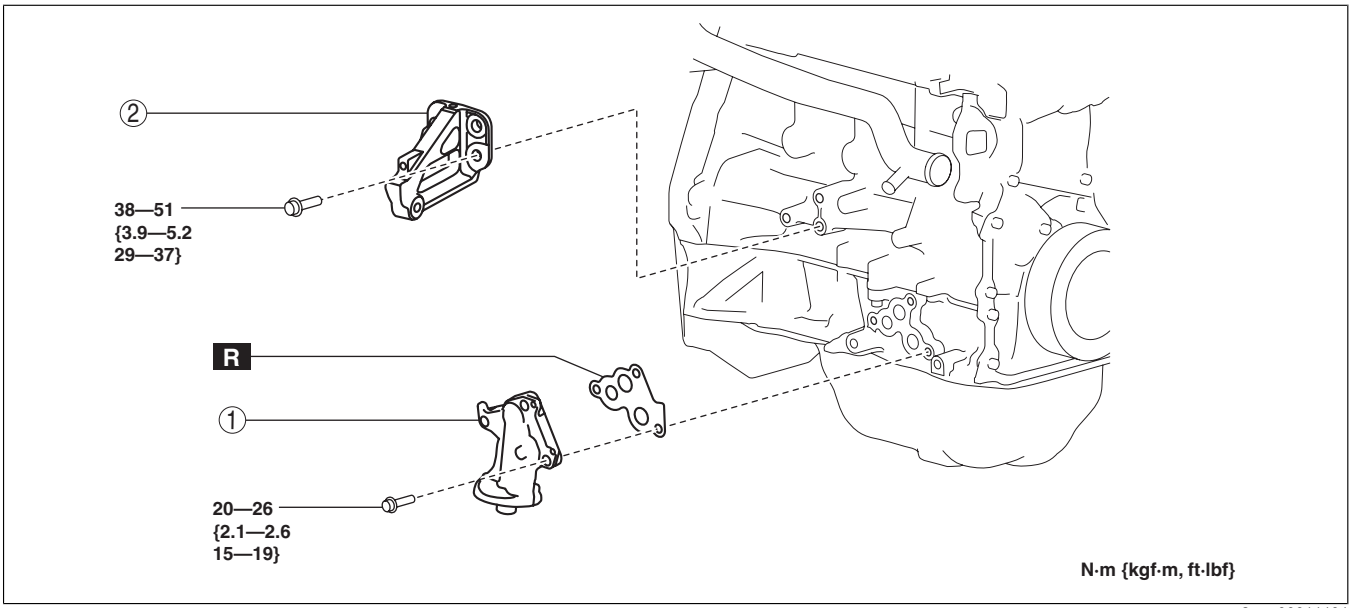
SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)



am2zzw00011100

1	Oil filter body (See Oil Filter Body Installation Note.)
2	Exhaust manifold bracket

SKYACTIV-G 1.5 (with 4-2-1 exhaust system)



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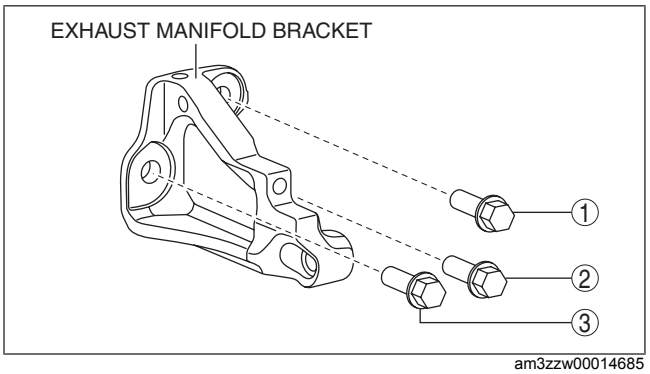
1	Oil filter body (See Oil Filter Body Installation Note.)
2	Exhaust manifold bracket (See Exhaust Manifold Bracket Installation Note (SKYACTIV-G 1.5 (With 4-2-1 Exhaust System)).)

Exhaust Manifold Bracket Installation Note (SKYACTIV-G 1.5 (With 4-2-1 Exhaust System))

1. Temporarily tighten the exhaust manifold bracket installation bolts.

2. Tighten the bolt in the order shown in the figure.

Tightening torque
38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}



am3zzw00014685

Oil Filter Body Installation Note

Note
- The tightening order of the three bolts is optional.

1. After tightening the three bolts, tighten the first tightened bolt to the specified tightening torque again.

Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}